

Week 1

9.3 星期二

1. (3) $\frac{3-4i}{4+3i} = -i$

2. (4) 设 $\theta_n = \theta_0 + 2n\pi$ ($n \in \mathbb{Z}$, $0 \leq \theta_0 < 2\pi$)

$\theta_0 = 0$ 时, $z = 0$, 辐角无意义

$\theta_0 \neq 0$ 时, $\arg z = \arctan \frac{\sin \theta_0}{1 - \cos \theta_0}$
 $= \arctan \left(\tan \frac{\theta_0}{2} \right)^{-1}$
 $= \frac{\pi - \theta_0}{2}$

$|z| = \sqrt{(1 - \cos \theta_0)^2 + \sin^2 \theta_0} = 2 \sin \frac{\theta_0}{2}$

2kπ 易错为 kπ

代入, 得: $z = 2 \sin \frac{\theta_0}{2} \cdot \exp \left(i \frac{\pi - \theta_0}{2} \right)$

$\text{Arg } z = \frac{\pi - \theta_0}{2} + 2k\pi$ ($k \in \mathbb{Z}$)

3. (3) $\sqrt[3]{1+i} = (\sqrt{2} \cdot e^{i\frac{\pi}{6}})^{\frac{1}{3}} = \sqrt[6]{2} \cdot \exp \left(i \left(\frac{\pi}{12} + \frac{2k\pi}{3} \right) \right)$ $k = 0, 1, 2$

忽略.

7. 记 $z = \sum_{k=1}^n \cos k\theta + i \sin k\theta$

$= e^{i\theta} \cdot \frac{1 - e^{in\theta}}{1 - e^{i\theta}}$

$= \frac{\cos \theta + i \sin \theta}{1 - \cos \theta - i \sin \theta} \cdot \frac{1}{(2 - 2 \cos \theta)} [(\cos \theta + \cos n\theta - \cos(n+1)\theta - 1) + i (\sin \theta + \sin n\theta - \sin(n+1)\theta)]$

12) (1) $= \text{Re } z = \frac{\cos \theta - 1 - 2 \sin \frac{2n+1}{2} \theta \cdot \sin \frac{\theta}{2}}{2 - 2 \cos \theta}$

$= -\frac{1}{2} + \frac{\sin \frac{2n+1}{2} \theta}{2 \sin \frac{\theta}{2}}$

(2) $= \text{Im } z = \frac{\sin \theta + \sin n\theta - \sin(n+1)\theta}{2 - 2 \cos \theta}$

$= \frac{1}{2} \cot \frac{\theta}{2} - \frac{\cos \frac{2n+1}{2} \theta}{2 \sin \frac{\theta}{2}}$

#

8. $|z_1 + z_2|^2 + |z_1 - z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$

$= 2(z_1 \bar{z}_1 + z_2 \bar{z}_2)$

$= 2(|z_1|^2 + |z_2|^2)$

平行四边形各边平方和等于对角线平方和.

9. ~~$z^n + a$~~ $\forall t \in \mathbb{C}$, $|t| \leq 1$ 若 $z^n = t$, 则

$$z = \sqrt[n]{|t|} e^{i \frac{1}{n} \arg t}, \text{ 此时 } |z| \leq 1$$

$\therefore z^n$ 可以是方向任意, 模长最大为 1 的复数.

$$\therefore |z^n + a|_{\max} = |z^n| + |a| = |a| + 1$$

此时, $\arg z^n = \arg a \Rightarrow z = e^{i \frac{1}{n} \arg a}$. 忽略取等条件

$$10. (1) \left| \frac{z-a}{1-\bar{a}z} \right| = \left| \frac{z-a}{\bar{z}(\bar{z}-\bar{a})} \right| = 1$$

$$(2) \left| \frac{z-a}{1-\bar{a}z} \right| = \frac{|z|^2 + |a|^2 - \bar{a}z - a\bar{z}}{1 + |az|^2 - \bar{a}z - a\bar{z}}$$

$$\therefore |z|^2 + |a|^2 - |az|^2 - 1 = (|a|^2 - 1)(1 - |z|^2) < 0$$

$$\therefore |z|^2 + |a|^2 - \bar{a}z - a\bar{z} < 1 + |az|^2 - \bar{a}z - a\bar{z}$$

$$\therefore \left| \frac{z-a}{1-\bar{a}z} \right| < 1$$

9.5 星期四

$$4. (2) z^3 = -i \Rightarrow z^3 = e^{-i \frac{\pi}{2}}$$

$$\Rightarrow z = \exp\left(i\left(-\frac{\pi}{6} + \frac{2\pi k}{3}\right)\right) \quad k = 0, 1, 2$$

$$6. (x+iy)^2 = (\sqrt{a+ib})^2$$

$$\Rightarrow \begin{cases} x^2 + y^2 = a \\ 2xy = b \end{cases} \Rightarrow x^2 = \frac{a + \sqrt{a^2 + b^2}}{2}, y^2 = \frac{-a + \sqrt{a^2 + b^2}}{2}$$

忽略 x, y 的符号.

代入验证得 x, y 有两解, $b > 0$ 时, x, y 同号; 反之, 异号.

$$16. (2) \text{ 通项为: } z_n = \frac{i^{n-1}}{n}$$

$$\therefore |z_n| = \frac{1}{n} \quad \lim_{n \rightarrow \infty} |z_n| = 0$$

忽略通项.

$$\therefore \text{有极限, } \lim_{n \rightarrow \infty} z_n = 0$$

18. (5) 以 $(\pm 1, 0)$ 为对称点的圆族, y 轴 ($\alpha = 1$)

19. (6) 非区域; 边界 $\{P \mid \sqrt{2} + x_p = -2, \sqrt{3} \leq |y_p| \leq 2\sqrt{2}\} \cup \{P \mid x_p = \frac{3}{2}, |y_p| \leq \frac{\sqrt{11}}{2}\} \cup \{P \mid r=2 \text{ 或 } r=3, -2 \leq x_p \leq \frac{3}{2}\}$; 图略.

(10) 是区域; 边界 $\{P \mid x_p = 0\} \setminus \{(-1, 0)\}$

21. (4) $z = t^2 + \frac{i}{t^2} \quad \therefore t^2 > 0$

曲线是 $y = \frac{1}{x} \quad (x > 0)$, 双曲线.

22. * $x^2 + 2x + y^2 = 1 \Rightarrow (x+1)^2 + y^2 = 2$

$\Rightarrow |z+1|^2 = 2$

$z = x + iy$

$\Rightarrow |z+1| = \sqrt{2}$

化简!!